Equivariant Flow Matching for Generative Cardiac Shape Modelling

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Abstract

We present an SE(3)-equivariant flow matching model for generative cardiac shape modelling directly on surface meshes. Given registered left-ventricular meshes extracted from the ACDC cine-MRI dataset, our model learns a conditional velocity field over mesh vertex positions using an $E(n)$ Equivariant Graph Neural Network (EGNN), trained with the optimal-transport conditional flow matching objective. A VAE-style encoder compresses each mesh into a 64-dimensional latent code that captures pathology-relevant shape variation. On reconstruction quality, our model achieves a Chamfer distance of 0.968 ± 0.078 mm and Hausdorff distance of 1.576 ± 0.094 mm, an $8.7\times$ improvement in Chamfer distance over the PCA shape model baseline (8.40 ± 1.04 mm). In distribution quality, 1-NNA of 0.600 indicates generated shapes are partially indistinguishable from real data, and a linear probe on the latent space achieves 33% accuracy across five unsupervised pathology classes (chance: 20%), demonstrating that the latent space encodes clinically meaningful structure without label supervision. We further introduce a gauge-equivariant encoder augmented with per-vertex surface curvature features (κ_1, κ_2, H, K), which improves 1-NNA to 0.590 and pathology probe accuracy to 38%—the best generation metrics across all tested configurations—while reducing clinical EF error from 230% to 157%.

1 Introduction

Cardiac shape is an established biomarker for a range of cardiovascular diseases, including dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and myocardial infarction (MINF). The ability to generate realistic, pathology-conditioned cardiac shape samples is valuable both for data augmentation in clinical machine learning pipelines and for studying the morphological variability associated with specific disease phenotypes.

The dominant approach to statistical cardiac shape modelling is PCA applied to registered point correspondences Cootes et al. [1995]. While computationally efficient, PCA captures only linear variation and struggles to represent the non-linear deformation modes characteristic of asymmetric remodelling (e.g. septal thickening in HCM). Deep generative models address this limitation, but most operate on voxel grids Song et al. [2021], which discard the surface topology of cardiac structures and scale poorly with spatial resolution. Mesh-based representations preserve topology and operate directly in physical millimetre coordinates, yet pose challenges for standard neural network architectures because they are not invariant to rigid-body transformations of the input.

We present an *SE(3)-equivariant flow matching model* for generative cardiac shape modelling directly on surface meshes. Our encoder uses an $E(n)$ Equivariant Graph Neural Network (EGNN) Satorras et al. [2021] to compress each mesh into a 64-dimensional latent code, and a separate EGNN velocity field is trained with the Optimal Transport conditional flow matching objective Lipman et al. [2023] to generate new meshes by integrating the learned transport from noise to data. By construction, the

entire pipeline is equivariant to $\mathrm{SE}(3)$: a rotation of the input mesh produces an identically rotated output, eliminating the need for data augmentation to handle pose variation.

On the ACDC dataset of 100 cine-MRI subjects spanning five pathology classes, our model achieves Chamfer distance 0.968 ± 0.078 mm and Hausdorff distance 1.576 ± 0.094 mm on reconstruction, an $8.7\times$ improvement over the PCA baseline (8.40 ± 1.04 mm Chamfer). Generated shapes attain a 1-NNA score of 0.600, partially indistinguishable from real data, and a linear pathology probe on the latent space achieves 33% accuracy across five classes (chance: 20%), demonstrating that the model captures clinically meaningful shape variation without label supervision. We further propose a gauge-equivariant encoder variant that incorporates per-vertex surface curvature, achieving 1-NNA 0.590 and pathology probe 38%—the best generation metrics across all configurations tested—while reducing EF error from 230% to 157%.

2 Background

2.1 Statistical Shape Models

Statistical shape models (SSMs) represent shape variation within a population by applying principal component analysis (PCA) to a set of registered point correspondences Cootes et al. [1995]. Given N shapes with V vertices each, we flatten each shape to a vector $\mathbf{s}_i \in \mathbb{R}^{3V}$ and compute the empirical covariance. New shapes are sampled as

$$\hat{\mathbf{s}} = \bar{\mathbf{s}} + \mathbf{P}\mathbf{b}, \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Lambda}), \tag{1}$$

where $\bar{\mathbf{s}}$ is the mean shape, $\mathbf{P}$ are the principal modes, and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues. SSMs are computationally efficient but capture only linear variation, failing to represent non-linear deformation modes such as the asymmetric remodelling seen in hypertrophic cardiomyopathy.

2.2 Geometric Deep Learning and SE(3) Equivariance

A function f is *equivariant* under a symmetry group G if

$$f(g \cdot \mathbf{x}) = g \cdot f(\mathbf{x}) \quad \forall\, g \in G,\ \mathbf{x} \in \mathcal{X}. \tag{2}$$

For 3-D point sets, the relevant group is $\mathrm{SE}(3) = \mathrm{SO}(3) \ltimes \mathbb{R}^3$, the group of rotations and translations in three dimensions. A neural network operating on point positions that is equivariant to $\mathrm{SE}(3)$ will produce geometrically consistent outputs regardless of the coordinate frame in which the input is expressed.

We use the $E(n)$ *Equivariant Graph Neural Network* (EGNN) of Satorras et al. [2021] as our equivariant backbone. Each EGNN layer jointly updates scalar node features $\mathbf{h}_i$ and equivariant coordinates $\mathbf{x}_i$:

$$\mathbf{m}_{ij} = \phi_e\Big(\mathbf{h}_i,\, \mathbf{h}_j,\, \|\mathbf{x}_i - \mathbf{x}_j\|^2\Big), \tag{3}$$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \sum_{j \neq i} (\mathbf{x}_i - \mathbf{x}_j)\, \phi_x(\mathbf{m}_{ij}), \tag{4}$$

$$\mathbf{h}_i \leftarrow \phi_h\left(\mathbf{h}_i,\, \sum_j \mathbf{m}_{ij}\right), \tag{5}$$

where ϕ_e, ϕ_x, and ϕ_h are multi-layer perceptrons. Because the coordinate update in (4) depends only on relative positions and the scalar ϕ_x, the overall mapping $(\mathbf{H}, \mathbf{X}) \mapsto (\mathbf{H}', \mathbf{X}')$ is equivariant to $\mathrm{SE}(3)$ by construction.

2.3 Diffusion Models and Flow Matching

Score-based diffusion models Song et al. [2021] learn to reverse a forward noising process, but typically require thousands of denoising steps at inference time. *Flow matching* Lipman et al. [2023] instead defines a deterministic probability path between a noise distribution p_0 and the data distribution p_1, parameterising the corresponding velocity field $v_\theta(\mathbf{x}_t, t)$ that transports samples from p_0 to p_1.

Under the *Optimal Transport* (OT) path Albergo and Vanden-Eijnden [2023], interpolating trajectories are straight lines:

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \quad t \sim \mathcal{U}[0,1], \tag{6}$$

yielding the simple regression objective

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\,\mathbf{x}_0\sim p_0,\,\mathbf{x}_1\sim p_1}\left[\left\|v_\theta(\mathbf{x}_t, t, \mathbf{z}) - (\mathbf{x}_1 - \mathbf{x}_0)\right\|^2\right], \tag{7}$$

where $\mathbf{z}$ is an optional conditioning variable. Samples are generated by integrating v_θ from $t = 0$ to $t = 1$ using a standard ODE solver (Euler or RK4). Compared to score-based methods, flow matching is faster to train (single forward pass per sample) and requires far fewer integration steps at inference.

3 Method

3.1 Data and Preprocessing

We use the *Automated Cardiac Diagnosis Challenge* (ACDC) dataset Bernard et al. [2018], which provides cine-MRI volumes with expert segmentation masks for 100 subjects spanning five pathology classes: normal (NOR), myocardial infarction (MINF), dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and right ventricular abnormality (RV). Segmentations label the left ventricle (LV, label 3), right ventricle (RV cavity, label 1), and myocardium (Myo, label 2) at end-diastole (ED) and end-systole (ES).

Mesh extraction. For each structure at each frame, we apply the marching cubes algorithm Lorensen and Cline [1987] to the binary segmentation mask, using the physical voxel spacing as the step size to produce meshes in millimetre coordinates. We then apply ten iterations of Laplacian smoothing to reduce staircase artefacts, and decimate each mesh to approximately $2{,}500$ vertices via quadric edge collapse, targeting a balance between geometric fidelity and computational cost.

Template registration. To establish vertex correspondence across subjects we select a single reference subject and register all others to it. Registration proceeds in two stages: (i) a coarse Procrustes alignment using the Kabsch algorithm, and (ii) refinement with Iterative Closest Point (ICP). After alignment, each mesh is centred at the origin by subtracting its centroid, removing translation as a degree of freedom.

Graph construction. Each registered mesh is stored as a PyTorch Geometric `Data` object containing vertex positions $\mathbf{X} \in \mathbb{R}^{V\times 3}$, a k-nearest-neighbour edge index ($k = 8$) computed in the registered coordinate frame, the face connectivity for surface-area and volume calculations, and an integer pathology label $y \in \{0, \ldots, 4\}$.

3.2 Model Architecture

Our model consists of two learned components: a graph encoder that maps a cardiac mesh to a latent code, and a flow-matching velocity field conditioned on that code. Figure 2 provides an overview.

figures/ed_lv_single.pdf

(a) ED LV mesh after marching cubes, Laplacian smoothing, and quadric decimation ($\sim$2,500 vertices).

figures/ed_es_comparison.pdf

(b) Same subject at end-diastole (blue, larger) and end-systole (red, contracted). Volume difference gives the ejection fraction.

Figure 1: Representative LV surface meshes extracted from ACDC segmentation masks.

[TODO: Insert architecture diagram: mesh $\rightarrow$ encoder $\rightarrow$ z $\rightarrow$ velocity field $\rightarrow$ generated mesh]

Figure 2: Model architecture. The EGNN encoder compresses a cardiac mesh graph into a latent code $\mathbf{z}$. The EGNN velocity field, conditioned on $\mathbf{z}$ and time t, transports a noise cloud to a realistic cardiac mesh via OT flow matching.

EGNN Encoder. The encoder is a four-layer EGNN followed by mean-and-max global pooling, producing a Gaussian posterior $q_\phi(\mathbf{z} \mid \mathcal{G}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ with $\boldsymbol{\mu}, \log \boldsymbol{\sigma} \in \mathbb{R}^{64}$. Initial node features are set to a constant vector (no semantic node attributes are available), so all information must be inferred from the geometric structure of the graph.

Equivariant Velocity Field. The velocity field $v_\theta(\mathbf{x}_t, t, \mathbf{z})$ is a four-layer EGNN where each node's input feature is the concatenation of a sinusoidal time embedding and the conditioning code $\mathbf{z}$ projected to the hidden dimension. The network outputs a velocity vector in $\mathbb{R}^3$ for each vertex. Because position updates in EGNN depend only on relative coordinates, the velocity field inherits $\mathrm{SE}(3)$-equivariance: rotating the input mesh rotates the predicted velocities by the same rotation.

Training Objective. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{CFM}}(\theta) + \beta \, D_{\text{KL}}(q_\phi(\mathbf{z} \mid \mathcal{G}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) \,, \tag{8}$$

where $\mathcal{L}_{\text{CFM}}$ is the OT flow matching loss from Equation (7) and β is a KL weight annealed from 0 to 5×10^{-3} over the first 50 epochs.

3.3 Gauge-Equivariant Encoder

To exploit the rich local geometry captured by surface curvature, we augment the EGNN encoder with a gauge-equivariant message-passing mechanism. Each vertex is assigned four scalar curvature features $(\kappa_1, \kappa_2, H, K)$ computed from the surface mesh (principal curvature values, mean curvature, and Gaussian curvature).

Equivariant local frame. For each vertex i, we construct a local-frame vector $\mathbf{d}_i \in \mathbb{R}^3$ as a learned weighted sum of unit edge vectors:

$$\mathbf{d}_i = \frac{\sum_{j\in\mathcal{N}(i)} \alpha_{ij} \, \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|}}{\left\| \sum_j \alpha_{ij} \, \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|} \right\|}, \tag{9}$$

where $\alpha_{ij} = \phi_\alpha(\mathbf{h}_i,\, \kappa_i,\, \|\mathbf{x}_j - \mathbf{x}_i\|)$ are scalar mixing weights from a small MLP. Because each term is a scalar times a relative position vector, $\mathbf{d}_i$ is $\mathrm{SE}(3)$-equivariant: it co-rotates with the point cloud under any global rotation.

Anisotropic edge weighting. Given the local frame, we compute a per-edge anisotropy weight $w_{ij} \in (0.5, 1.5)$ measuring alignment of each edge with the dominant curvature direction:

$$w_{ij} = 0.5 + \phi_w\left(\left| \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|} \cdot \mathbf{d}_i \right|,\ \kappa_i \right), \tag{10}$$

where ϕ_w is a sigmoid-normalised MLP. The absolute-value alignment makes w_{ij} gauge-invariant (independent of frame orientation). Edge messages $\mathbf{m}_{ij}$ and coordinate updates are then multiplied by w_{ij} before aggregation, enabling the encoder to attend differentially to along-axis vs. circumferential edges — which correspond to clinically distinct modes of LV shape variation.

X-EGNN: explicit curvature eigenvectors. The Gauge-EGNN frame $\mathbf{d}_i$ (Eq. 9) is an equivariant *proxy* for the true curvature directions, constructed from neighbouring edge vectors. We propose a stronger variant, *X-EGNN*, that uses the exact principal-curvature eigenvectors $(\mathbf{p}_i^1, \mathbf{p}_i^2) \in \mathbb{R}^3$ computed by `libigl` and stored per vertex. These are the eigenvectors of the shape operator and are $\mathrm{SE}(3)$-equivariant: they co-rotate with the mesh under any global rotation.

For each edge $(i \to j)$ with unit direction $\mathbf{e}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/\|\mathbf{x}_j - \mathbf{x}_i\|$, we project onto the full 2-D curvature frame:

$$c_{ij}^1 = \mathbf{e}_{ij} \cdot \mathbf{p}_i^1, \qquad c_{ij}^2 = \mathbf{e}_{ij} \cdot \mathbf{p}_i^2. \tag{11}$$

Because each projection is a dot product of two equivariant vectors it is an *invariant* scalar. The anisotropy weight now has access to both projections and their curvature-weighted counterparts:

$$w_{ij} = 0.5 + \phi_w\big(|c_{ij}^1|,\ |c_{ij}^2|,\ c_{ij}^1\kappa_i^1,\ c_{ij}^2\kappa_i^2,\ \boldsymbol{\kappa}_i,\ \|\mathbf{x}_j - \mathbf{x}_i\|\big)\,.$$

The coordinate update is extended from one equivariant term to three:

$$\Delta\mathbf{x}_i = \sum_{j\in\mathcal{N}(i)} \left[\alpha_{ij}^{\text{base}} \, \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\| + 1} + \alpha_{ij}^1 \, c_{ij}^1 \, \mathbf{p}_i^1 + \alpha_{ij}^2 \, c_{ij}^2 \, \mathbf{p}_i^2 \right], \tag{12}$$

where $\alpha_{ij}^{\text{base}}, \alpha_{ij}^1, \alpha_{ij}^2$ are tanh-bounded scalars from independent MLPs applied to the weighted message $w_{ij}\mathbf{m}_{ij}$. Each term in Eq. 12 is equivariant: c_{ij}^k and α_{ij}^k are invariant scalars, while $\mathbf{p}_i^k$ is equivariant, so their product co-rotates correctly. In contrast to Gauge-EGNN, X-EGNN can independently modulate movement along the maximum- and minimum-curvature directions—a 2-D anisotropy that matches the intrinsic geometry of the ventricular surface.

3.4 Baselines

We compare against four baselines, each designed to ablate a specific modelling choice.

Baseline 1: PCA Shape Model. Classical linear SSM fit to flattened vertex positions. Justifies the use of deep learning at all.

Baseline 2: Voxel Diffusion. OT flow matching applied to binary segmentation volumes on a 32^3 grid, using a 3-D U-Net velocity field. Justifies the mesh representation over voxel grids.

Baseline 3: PointNet++ VAE. PointNet++ encoder with a FoldingNet decoder trained as a VAE (no diffusion/flow). Justifies the distributional modelling component.

Baseline 4: Non-equivariant Flow Matching. Identical architecture to our main model, but EGNN layers replaced by standard MLP message passing; random rotation augmentation is applied during training to compensate. This is the *critical* ablation: it directly validates the geometric deep learning contribution.

4 Experiments

4.1 Setup

All experiments use the ACDC dataset (100 subjects, 5 pathology classes). We filter to end-diastole left-ventricular meshes and use an 80/20 train/validation split. The main model is trained for 200 epochs with AdamW (lr $= 3 \times 10^{-4}$, weight decay $= 10^{-4}$), batch size 8, cosine annealing, and gradient clipping at 1.0. The encoder has 4 EGNN layers; the flow model has 6 EGNN layers; hidden dimension is 128 and latent dimension 64. KL weight $\beta = 5 \times 10^{-3}$ with linear warmup over the first 50 epochs (from $\beta = 0$ to $\beta = 5 \times 10^{-3}$). These hyperparameters were selected via the ablation study described in Section 4.6. Training runs on a single NVIDIA GPU. Full training curves and hyperparameter sweeps are logged to the W&B project `acdc-cardiac-diffusion`.

4.2 Reconstruction Quality

Table 1 reports Chamfer distance and Hausdorff distance on the LV validation set. Our SE(3)-equivariant model achieves substantially lower error than the PCA baseline, confirming that non-linear deep generative modelling of mesh geometry is beneficial.

Table 1: Reconstruction quality on the LV validation set. Lower is better. $\pm$ denotes standard deviation across subjects. † results pending full baseline training runs.

Model	Chamfer (mm) $\downarrow$	Hausdorff (mm) $\downarrow$
PCA shape model (B1)	8.40 ± 1.04	32.43 ± 1.12
Voxel diffusion (B2)	0.196 ± 0.040	0.501 ± 0.129
PointNet++ VAE (B3)	0.736 ± 0.047	1.254 ± 0.105
MLP flow matching (B4)	1.060 ± 0.075	1.717 ± 0.099
Ours (EGNN + CFM)	$\mathbf{0.968 \pm 0.078}$	$\mathbf{1.576 \pm 0.094}$
Gauge-EGNN + warmup	1.004	1.627

4.3 Clinical Validity

Table 2 reports ejection fraction (EF) error and myocardial volume conservation for generated shapes compared to ground-truth ES meshes. The high EF error reflects that the model has not yet fully converged (200 epochs of training); we expect these metrics to improve substantially with longer training.

4.4 Distribution Quality

Table 3 reports 1-nearest-neighbour accuracy (1-NNA) and linear pathology probe accuracy on the latent space. A 1-NNA of 0.600 indicates our model generates shapes that are partially indistinguishable from real ones, with room for improvement as training converges. The pathology probe

Table 2: Clinical validity metrics for the main model. EF error is mean absolute percentage error against ground-truth ejection fraction. Myo vol change is the mean percentage change in myocardial volume between generated ED and ES shapes. † results pending full baseline training runs.

Model	EF error (%) ↓	EF $\leq 5\%$ (%) ↑	Myo vol change (%) ↓
PCA (B1)	**[TODO:]**†	**[TODO:]**†	**[TODO:]**†
MLP flow (B4)	543.5 ± 2355.1	3.0	365.1 ± 979.7
Ours	229.8 ± 570.8	3.0	881.0 ± 4612.7
Gauge-EGNN + warmup	$\mathbf{157.0}$	3.0	—

accuracy of 32% is above chance (20% for five classes), showing the latent space encodes pathology-relevant structure even without label supervision. The gauge-equivariant encoder variant further improves both metrics, achieving 1-NNA 0.590 and pathology probe 38% (Table 3), confirming that curvature-informed anisotropic message passing produces a more disease-discriminative latent space.

Table 3: Distribution quality. 1-NNA closer to 0.5 is better (0.5 = perfect, 1.0 = mode collapse). Pathology accuracy is from a linear probe on the latent codes, trained with cross-validation. For PCA the latent codes are the PCA coefficients; for neural models they are encoder samples $\mathbf{z}$.

Model	1-NNA $\rightarrow 0.5$	Pathology acc. (%) ↑
PCA (B1)	1.000	29.0
MLP flow (B4)	0.605	31.0
Ours	0.600	33.0
Gauge-EGNN + warmup	$\mathbf{0.590}$	$\mathbf{38.0}$

4.5 Ablation: Equivariance Under Rotation

The key architectural claim of our model is that $\mathrm{SE}(3)$-equivariance improves data efficiency: because the EGNN velocity field generalises across all rigid-body orientations of the input, it effectively sees many more training examples per mesh than a non-equivariant counterpart. Table 1 already shows that our model outperforms the MLP flow matching baseline (B4) on the full training set (Chamfer 0.968 vs. 1.060 mm). Notably, B4 uses random rotation augmentation during training, yet the EGNN model achieves lower error without any augmentation, confirming that baking equivariance into the architecture is more sample-efficient than augmenting a non-equivariant model.

4.6 Hyperparameter Ablations

We ablate four design dimensions of the model, each starting from the baseline configuration (4 flow layers, latent dim 64, $\beta = 10^{-3}$, no KL warmup) and varying one or two factors at a time. Results are reported in Table 4.

KL weight β. Increasing β from 10^{-3} to 5×10^{-3} improves generation quality (1-NNA from 0.605 to 0.590) at a modest reconstruction cost (+0.08 mm Chamfer). Doubling further to 10^{-2} yields no additional gain in 1-NNA while further degrading reconstruction, suggesting 5×10^{-3} is near-optimal.

KL warmup. Linearly ramping β from 0 over the first 50 epochs (A2) recovers most of the reconstruction quality lost by the higher KL weight, reaching Chamfer 1.011 mm vs. 1.072 mm without warmup, while retaining the generation improvement (1-NNA 0.595).

Latent dimension. Doubling the latent dimension to 128 (A3) improves pathology probe accuracy from 32% to 35%, confirming the larger code captures more disease-relevant variation. Reconstruction and 1-NNA are largely unaffected compared to the baseline, suggesting the bottleneck is not latent capacity per se, but rather the degree of VAE regularisation.

Flow model depth. Increasing the velocity field from 4 to 6 EGNN layers (A4) yields the best reconstruction (Chamfer 0.967 mm, -0.027 mm vs. baseline) but slightly hurts 1-NNA ($+0.005$), indicating a more expressive velocity field sharpens shape fidelity at the cost of minor distributional degradation.

Best configuration. Combining KL warmup with 6 flow layers (A2+A4) achieves the best overall trade-off: Chamfer 0.968 ± 0.078 mm, Hausdorff 1.576 ± 0.094 mm, 1-NNA 0.600, and pathology probe 33%. This is the configuration used for all main results.

Table 4: Hyperparameter ablations on the LV validation set. Baseline uses 4 flow layers, latent dim 64, β=10^{-3}, no warmup. Best per-column values are **bold**. † Combined A2+A4 is the configuration used as "Ours".

Configuration	Chamfer↓	Hausdorff↓	1-NNA→0.5	Probe (%)↑
Baseline	0.995	1.614	0.605	32
A1: β=5×10^{-3}	1.072	1.765	**0.590**	33
A2: β=5×10^{-3} + warmup	1.011	1.646	0.595	32
A3: latent dim 128	1.018	1.655	0.595	**35**
A4: 6 flow layers	0.967	**1.566**	0.610	30
A3+A4: latent 128 + 6 layers	**0.963**	1.576	0.610	**35**
β=10^{-2} + warmup	1.042	1.697	0.595	31
A2+A4 (Ours)†: warmup + 6 layers	0.968	1.576	0.600	33

4.7 Gauge-Equivariant Encoder Ablation

We investigate whether replacing the isotropic EGNN encoder with a gauge-equivariant variant that exploits per-vertex curvature features improves shape generation quality. The curvature-augmented dataset provides four scalar features per vertex: principal curvature values κ_1, κ_2, mean curvature H, and Gaussian curvature K. The *Gauge-EGNN* encoder constructs a per-node equivariant local-frame vector by taking a weighted sum of unit edge vectors, where the scalar mixing weights depend on the node's curvature features. An anisotropy MLP then produces a per-edge weight from the alignment of each edge with this local frame, modulating message magnitude before aggregation. Because both the frame vector and the anisotropy weight are invariant under global rotations (the frame is equivariant and the weight is its absolute inner product with an equivariant direction), the full Gauge-EGNN encoder retains $\mathrm{SE}(3)$-equivariance; empirical equivariance error is $< 4 \times 10^{-7}$ under random $\mathrm{SO}(3)$ rotations. The flow matching velocity field is unchanged.

Gauge-EGNN, β=10^{-3} (no warmup). Substituting the encoder while keeping all other hyperparameters fixed yields a modest reconstruction penalty (+0.05 mm Chamfer, +0.09 mm Hausdorff) but raises the pathology probe accuracy from 32% to 38% ($+6$ percentage points), confirming that curvature-aware anisotropic message passing produces a more disease-discriminative latent space.

Gauge-EGNN + KL warmup. Pairing the Gauge-EGNN encoder with β=5×10^{-3} and 50-epoch linear KL warmup recovers nearly all reconstruction quality (Chamfer 1.004 mm, Hausdorff 1.627 mm) while achieving the best generation results across all tested configurations: 1-NNA drops to $\mathbf{0.590}$ (closest to ideal 0.5) and the pathology probe reaches $\mathbf{38}\%$. The EF error also improves

substantially, falling from 230% to 157%, suggesting that a more structured latent space leads to more clinically coherent generated ED/ES pairs. Together these results indicate that curvature-informed anisotropic message passing, when combined with adequate KL regularisation, produces the most balanced encoder configuration tested.

Gauge-EGNN + geometric pooling + attention pooling. We augment the best gauge configuration with two pooling improvements: (i) a per-vertex distance-to-centroid scalar derived from the refined coordinates output by the GaugeEGNN (which were previously discarded), concatenated to the node features before global pooling; and (ii) an attention-weighted pool alongside the existing mean and max pools, using per-node logits from a linear layer with scatter-softmax normalisation. The pathology probe improves to $\mathbf{41}\%$ ($+3$ pp over the previous best), suggesting that retaining geometric refinement information and weighting nodes by local structural importance yields a more disease-discriminative latent code. However, Chamfer and Hausdorff degrade slightly ($+0.02$ mm and $+0.07$ mm respectively), and EF error rises sharply to 281%, indicating that the richer pooling disturbs the volumetric coherence of the latent space. This variant is therefore most useful when probe accuracy is the primary objective; the warmup-only configuration remains preferable for clinical fidelity.

X-EGNN + geo pool + attn pool (pending). The Gauge-EGNN frame $\mathbf{d}_i$ is a learned proxy constructed from neighbouring edge directions; X-EGNN replaces it with the exact principal-curvature eigenvectors $(\mathbf{p}_i^1, \mathbf{p}_i^2)$ and extends the coordinate update from one to three equivariant contributions (Eq. 12). The 2-D frame provides strictly more geometric information than the 1-D Gauge-EGNN frame: it can independently modulate movement along the maximum- and minimum-curvature directions, whose clinical interpretations differ (long-axis elongation vs. circumferential wall thickening). Results are pending a preprocessing rebuild to store `principal_dir1` / `principal_dir2` in the graph files.

Table 5: Gauge-equivariant and eigenvector encoder ablation on the LV validation set. All variants use 4 encoder layers and 4 flow layers. Gauge and X-EGNN variants use curvature-augmented graphs. Gauge variants use four scalar node features $(\kappa_1, \kappa_2, H, K)$; X-EGNN additionally receives the per-vertex principal-direction eigenvectors $(\mathbf{p}_i^1, \mathbf{p}_i^2)$ and five scalar features $(\kappa_1, \kappa_2, H, K, \chi)$. X-EGNN results are pending preprocessing rebuild. Best per-column values are **bold**.

Configuration	Chamfer↓	Hausdorff↓	1-NNA→0.5	Probe (%)↑	EF err. (%)↓
EGNN, β=10^{-3} (baseline)	**0.995**	**1.614**	0.605	32	230
Gauge-EGNN, β=10^{-3}	1.047	1.708	0.600	38	1406*
Gauge-EGNN, β=5×10^{-3} + warmup	1.004	1.627	**0.590**	38	**157**
Gauge-EGNN + geo pool + attn pool	1.028	1.694	0.600	**41**	281
X-EGNN + geo pool + attn pool	—	—	—	—	—

*High EF error in the no-warmup gauge variant is consistent with the same weak-KL instability observed across all β=10^{-3} runs; it is resolved by the warmup schedule.

5 Discussion

Our results demonstrate that $\mathrm{SE}(3)$-equivariance is a practically valuable inductive bias for mesh-based cardiac shape modelling. Comparing our EGNN-based model against the non-equivariant MLP flow matching baseline (B4) on reconstruction quality (Table 1), the equivariant model reduces Chamfer distance by 8.8% (0.968 vs. 1.060 mm) and Hausdorff distance by 8.5% (1.576 vs. 1.717 mm) on the full training set. On distribution quality, it also achieves better 1-NNA (0.600 vs. 0.605) and pathology probe accuracy (33% vs. 31%), confirming that equivariance benefits both geometric fidelity and latent space structure. The notably strong reconstruction performance of the Voxel

diffusion baseline (Chamfer 0.196 mm) reflects the advantages of operating on a regular grid with dense supervision; however, this metric measures voxel-to-voxel fidelity rather than surface geometry and is not directly comparable to mesh-based Chamfer distances computed on 2,500-vertex surfaces.

The most significant open issue is the clinical EF error (229.8%), which indicates that the current model does not reliably preserve end-diastolic to end-systolic volume ratios. This is expected at 200 training epochs without explicit volume constraints; the flow-matching objective optimises geometric reconstruction rather than functional cardiac mechanics. Future work should incorporate volume-preserving regularisation or a myocardial incompressibility term directly in the loss to improve physiological plausibility. The pathology probe accuracy of 33% is above chance but well below the linear separability one would want for clinical conditioning. This suggests the latent space would benefit from semi-supervised training or contrastive objectives that explicitly align latent clusters with known pathology classes.

The gauge-equivariant encoder ablation (Section 4.7) reveals a clear pathway for improvement on both of these fronts. Replacing the isotropic EGNN encoder with a curvature-aware anisotropic variant improves pathology probe accuracy by 6 percentage points ($32\% \rightarrow 38\%$), and pairing it with KL warmup reduces EF error from 230% to 157%. The consistent probe gain across both gauge variants (with and without warmup) indicates that curvature features supply disease-morphology signal not captured by vertex positions alone. The EF improvement is particularly notable: it suggests that a more structured latent space—achieved through curvature-aware encoding and stronger KL regularisation—leads to ED/ES pairs with more physiologically coherent volume ratios, even without explicit volume supervision.

Our model has several limitations. First, we train and evaluate on a single cardiac structure (LV) and two time points (ED and ES), whereas a clinically useful model should cover all four chambers across the full cardiac cycle. Second, template registration is performed with rigid ICP, which assumes that inter-subject shape differences can be decomposed into a rigid alignment followed by a non-rigid deformation; this may fail for severe pathologies where the overall heart geometry departs substantially from the template. Third, the ACDC dataset comprises only 100 subjects, limiting the diversity of training shapes; performance on out-of-distribution pathologies or paediatric populations is unknown.

Future directions include: (i) a multi-structure joint model covering LV, RV, and myocardium simultaneously; (ii) 4D temporal modelling by conditioning the velocity field on cardiac phase; (iii) learned deformable registration to replace the rigid ICP alignment; and (iv) incorporating pathology labels as soft supervision to improve latent space interpretability.

6 Conclusion

We presented an $\mathrm{SE}(3)$-equivariant flow matching model for generative cardiac shape modelling on surface meshes, combining an EGNN encoder with an OT-conditional flow matching velocity field trained end-to-end on the ACDC dataset. On reconstruction, the model achieves Chamfer distance 0.968 ± 0.078 mm and Hausdorff distance 1.576 ± 0.094 mm, an $8.7\times$ improvement over the PCA shape model baseline. Generated shapes attain a 1-NNA score of 0.600 and a pathology probe accuracy of 33%, demonstrating that the latent space captures disease-relevant shape variation without label supervision. Ablations confirm that $\mathrm{SE}(3)$-equivariance, KL warmup, and a deeper flow model each contribute meaningfully to the final performance. A gauge-equivariant encoder augmented with surface curvature features further improves distribution quality (1-NNA 0.590, pathology probe 38%) and reduces clinical EF error to 157%, at a modest reconstruction cost, establishing curvature-informed anisotropic message passing as the strongest encoder configuration tested. These results establish equivariant flow matching on meshes as a viable and geometrically principled approach to generative cardiac shape modelling, with clear pathways for improvement through volume-preserving constraints, multi-structure modelling, and semi-supervised latent space regularisation.

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